

TECHBIG SOLUTIONS

SAP S/4HANA – Production Planning (PP)

Complete Professional Training Syllabus

Ambattur, Chennai

Course Objective

A comprehensive functional curriculum covering SAP S/4HANA Production Planning from organizational structure and master data through demand management, MRP/MRP Live, capacity planning, production orders, confirmations, goods movements, repetitive manufacturing, Kanban, integration, configuration, monitoring and real-time implementation/support scenarios.

The scope follows current SAP S/4HANA learning themes including supply-chain planning, demand management, MRP, capacity evaluation and embedded PP/DS. SAP Learning also identifies make-to-stock and make-to-order planning, MRP Live, predictive MRP and demand-driven replenishment as S/4HANA planning topics. ■cite■turn0search1■turn0search0■

1. SAP S/4HANA Manufacturing & PP Overview

- S/4HANA manufacturing architecture and PP positioning
- Production Planning versus execution
- End-to-end Plan-to-Produce process
- Organizational structure: company code, plant, storage location, MRP area
- Manufacturing scenarios and process variants
- SAP Fiori and role-based navigation overview
- Integration with MM, SD, CO, QM, EWM/WM and PM

2. PP Master Data

- Material master and MRP views
- MRP type, lot sizing and procurement type
- Work centers and capacity data
- BOMs: usage, alternatives and validity
- Routings and operation sequences
- Production versions
- Task lists and formulas
- Production resources/tools overview
- PIR-related master data
- Master-data consistency and common errors

3. Bill of Material (BOM)

- BOM creation/change/display
- BOM structures and item categories
- Alternative BOMs
- BOM usage and status
- Validity dates and engineering change concepts
- BOM explosion in MRP and production orders
- BOM transfer and integration considerations

4. Work Centers & Capacity

- Work center structure and basic data
- Scheduling, costing and capacity data
- Capacity categories
- Available capacity and formulas
- Individual versus collective capacities
- Capacity requirements
- Capacity evaluation and overload analysis
- Shift and operating-time concepts

5. Routings & Production Versions

- Routing creation and maintenance
- Operations, sequences and relationships
- Control keys
- Standard values and formulas
- Scheduling parameters
- Operation components
- Production version selection
- Routing/BOM combination
- Alternative production methods

6. Demand Management & Planning Strategies

- Planned Independent Requirements (PIRs)
- Demand management concepts
- Consumption-based planning
- Make-to-stock
- Make-to-order
- Planning with final assembly
- Planning without final assembly
- Strategy groups and requirements types
- Customer requirements and demand sources
- Planning strategy selection by business scenario

7. MRP / MRP Live in S/4HANA

- MRP fundamentals and planning logic
- MRP Live overview and differences from classic MRP
- Planning run and scope
- Net requirements calculation
- BOM explosion
- Procurement proposals
- Planned orders and purchase requisitions
- Scheduling and lead times
- Exception messages
- Planning file and planning-relevant parameters
- MRP controller and MRP areas
- Multi-level planning

8. Lot Sizing & Procurement Planning

- Static and dynamic lot-sizing concepts
- Minimum/maximum lot size
- Rounding values
- Safety stock

- Reorder point planning overview
- In-house production
- External procurement
- Special procurement keys
- Subcontracting planning overview
- Stock transfer planning overview

9. Capacity Planning & Evaluation

- Capacity requirements from planned/production orders
- Capacity evaluation
- Work center load analysis
- Capacity leveling concepts
- Finite and infinite planning concepts
- Capacity overload resolution
- Scheduling impact on capacity
- Capacity planning reports and Fiori apps

10. Planned Orders

- Planned-order creation and display
- Conversion of planned order to production order
- Planned-order scheduling
- Component requirements
- Capacity requirements
- Firming planned orders
- Mass processing
- Planning proposal review
- Exception handling

11. Production Orders

- Production order structure
- Order types and parameters
- Create/release/change/display production orders
- BOM and routing explosion
- Scheduling and dates
- Availability checks
- Component staging
- Capacity checks
- Order release and execution controls
- Order printing and shop-floor documents

12. Production Order Execution

- Goods issue to production
- Backflush concepts
- Confirmation of operations
- Partial and final confirmation
- Yield, scrap and rework
- Time confirmations
- Automatic goods movements
- Error handling in confirmations
- Production order status management
- Order completion and technical completion

13. Goods Movements & Inventory Integration

- 261 goods issue and reversal
- 101 goods receipt and reversal
- Component reservations
- Batch-managed materials
- Serial-number considerations
- Storage locations and staging
- MM-PP integration
- Stock reconciliation
- Common movement errors

14. Production Order Settlement & Cost Integration

- Product costing overview
- Planned versus actual costs
- Activity confirmations and CO integration
- WIP overview
- Variance calculation
- Settlement concepts
- Production variances
- Period-end manufacturing activities
- Integration with Controlling

15. Repetitive Manufacturing (REM)

- REM business scenarios
- Repetitive manufacturing master data
- Rate-based planning
- Production lines
- Reporting points
- Backflush
- Goods movements

- REM planning and execution
- REM versus discrete manufacturing

16. Kanban

- Kanban concepts and replenishment
- Control cycles
- Kanban containers
- Status-driven replenishment
- Internal production Kanban
- External procurement Kanban
- Goods movements and replenishment
- Kanban monitoring

17. Production Planning with Sales & Distribution Integration

- Sales order demand
- Make-to-order planning
- Delivery requirements
- Availability and production planning
- SD-PP integration flow
- Customer-specific production scenarios
- Order and requirement tracking

18. PP Integration with MM, QM, PM and EWM/WM

- MM procurement and inventory integration
- QM inspection during production and GR
- PM maintenance-resource considerations
- Warehouse staging and goods issue
- EWM integration overview
- Production supply areas
- Stock transfer and staging concepts
- Cross-functional troubleshooting

19. Embedded SAP S/4HANA PP/DS – Advanced Planning

- Embedded PP/DS concept
- Master data requirements
- Product view
- Alert Monitor
- Planning strategies
- Heuristics
- Detailed Scheduling
- Finite capacity planning

- Resource scheduling
- Pegging and alerts
- Optimization concepts
- Interaction between S/4HANA planning and embedded PP/DS

20. Predictive MRP, Demand-Driven & Advanced Planning Overview

- Predictive MRP (pMRP) concept
- Simulation and planning evaluation
- Demand-driven replenishment overview
- Multi-level planning concepts
- When to use advanced planning
- Planning exceptions and alerts

21. Configuration / SPRO Curriculum

- Enterprise structure and plant settings
- Define MRP controllers and parameters
- MRP types and lot-sizing procedures
- Number ranges and order types
- Production scheduling parameters
- Order-type-dependent parameters
- Availability checking
- Scheduling profiles
- Confirmation parameters
- Goods movement settings
- Production versions
- Capacity configuration
- Planning strategies and requirements types
- Integration settings
- Testing and transport considerations

22. Monitoring, Reporting & Fiori

- MRP results and exception monitoring
- Stock/requirements list concepts
- MRP list where applicable
- Production order information system
- Capacity evaluation
- Order progress monitoring
- Material availability monitoring
- Fiori apps for production planning and execution
- Exception-based management

23. Production Support & Troubleshooting

- MRP not creating planned orders
- Incorrect BOM/routing selection
- Production version issues
- Capacity overload
- Availability check failures
- Goods movement errors
- Backflush errors
- Confirmation errors
- Batch determination issues
- Order status problems
- Incorrect component quantities
- Settlement/period-end issues
- Root-cause analysis, testing and incident documentation

24. End-to-End Real-Time Projects

- Project 1: Make-to-stock – PIR → MRP → planned order → production order → GI → confirmation → GR
- Project 2: Make-to-order – sales order → planning → production → delivery integration
- Project 3: Multi-level BOM planning and execution
- Project 4: Capacity-constrained production
- Project 5: Batch-managed production
- Project 6: Repetitive manufacturing
- Project 7: Kanban replenishment
- Project 8: PP/DS planning scenario
- Project 9: Production support incident and root-cause analysis

25. Interview & Certification Preparation

- PP configuration interview questions
- MRP and MRP Live scenarios
- Master-data troubleshooting
- Production-order scenarios
- Integration questions
- Real-time implementation cases
- Support-ticket scenarios
- Mock interviews
- Practical assessment and project presentation

Suggested Training Roadmap

Phase	Coverage	Outcome
1	Foundation & Master Data	Understand S/4HANA PP architecture and create core master data
2	Demand & MRP	Run planning, analyze results and resolve exceptions
3	Capacity & Orders	Plan capacity and execute production orders
4	Execution	GI, confirmation, GR, batch and shop-floor scenarios
5	Integration	MM, SD, CO, QM and warehouse integration
6	Advanced	REM, Kanban and embedded PP/DS overview
7	Configuration	Configure core PP processes and test end-to-end
8	Project & Support	Complete implementation case and production-support scenarios

SAP Reference Alignment

SAP Learning's current S/4HANA Production Planning material covers supply-chain planning, demand management, MRP/MRP Live, capacity evaluation, make-to-stock/make-to-order and advanced PP/DS. SAP also describes embedded PP/DS as integrated into S/4HANA for finite planning and detailed scheduling.

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